POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Architecture

Semester: Fall

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Differentiate Computer Organization and architecture. Explain the various addressing modes. 8
   b) Define RTL. Explain about BUS and Memory transferin RTL. 7
2. a) Explain different types of registers used in computer system. What is use of register in computer system? 7
   b) Perform multiplication -13*7 using Booth's Algorithm. 8
3. a) What are the various number representation method in computer system? 7
   b) Explain block diagram of micro programmed control organization and write down its advantages. 8
4. a) Explain associative and set-associative mapping in cache memory? 8
   b) Define an interrupt with its classes in detail. Also explain about interrupt driven I/O. 7
5. a) What is pipelining hazards? Explain structural and data hazard and its solution 7
   b) What is cache coherence problem? Explain software and hardware solution of cache coherence problem. 8
6. a) Write about parallelism in computer system. Explain about symmetric multiprocessor in detail. 7
   b) Explain about the various Hardware Performance issues in Multicore Computers. 8

7. Write short notes on: **(Any two)**
   a) Shift Micro operations
   b) Magnetic tape and disk
   c) Differentiate between RISC and CISC.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Architecture

Semester: Fall

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define Computer Architecture and Computer Organization. What is HDL and what are its advantages? 6
   b) What is addressing mode? Explain about different types of addressing modes. 9
2. a) What is CPU organization? Explain register organization in details. 8
   b) What is ALU? Design a 2-bit ALU that can perform AND, OR & ADD operations. 7
3. a) What is Booths algorithm in computer Architecture? Explain 8
   b) What is associative memory? How read write operation is performed in associative memory? 7
4. a) Draw the block diagram of micro programmed control unit and explain it. 6
   b) What the drawbacks are of interrupt driven and programmed I/O? How DMA overcomes it? Explain. 9
5. a) Explain DMA transfer in a Computer system with necessary diagram. 6
   b) Compare RISC and CISC Architecture. Describe Flynn's classification. 9
6. a) Show that the speedup factor for pipelined processor is equal to the number of stages in a pipeline. 7
   b) What is cache mapping? Explain cache replacement algorithm with a suitable example. 8

7. Write short notes on: **(Any two)**

a) RAID

b) Hardware performance issues in multicore systems

c) Pipelining Conflicts

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Architecture

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) What are the major differences between Computer Architecture and Computer Organization? Explain about different categories of micro operations. 6

b) What is addressing mode? Write Codes to evaluate X = (A-B*F)*C+D/E using three, two, one and zero address instruction formats. 9

a) Define data path. What are the design principles for modern CPU system? 7

b) Explain Booth's algorithm with example of your own. 8

a) Draw the circuit for BCD adder. Explain floating point representation. 7

b) Distinguish hardwired control unit and micro programmed control unit with suitable example of each. 8

a) Explain about Associative mapping technique. Why replacement algorithm is required in associative mapping technique? 7

b) Describe the drawbacks of Programmed I/O and interrupt driven I/O and explain how DMA overcomes these drawbacks. 8

a) What is hit ratio in cache memory? Mention the advantages and disadvantages of various cache mapping techniques. 8

b) Differentiate between RISC and CISC pipelining. 7

a) Differentiate between multi computer and multiprocessor system. List out various multiprocessor interconnection structure. How does the cache coherence problem arise in multiprocessor system? Explain with example. 8

b) Show that the speedup factor for pipelined processor is equal to the number of stages in a pipeline. 7

7. Write short notes on: **(Any two)**

a) Vector Processor

b) Role of I/O interface in Computer system

c) RAID

POKHARA UNIVERSITY

Semester: Fall

Level: Bachelor
Programme: BE
Course: Computer Architecture

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Discuss the term Computer Architecture. Explain IAS computer architecture with detailed diagram. 8

b) Define addressing mode. Explain different addressing modes with appropriate figures and example of each. 7

a) Define RTL. How data can be transferred between registers. Explain. 8

b) Define Instruction cycle. What are the different states in an instruction cycle? Also explain fetch cycle, decode cycle, execute cycle and indirect cycle. 8

a) Draw the flowchart for Booth's Algorithm. Use the algorithm for binary multiplication of (-12) and (-9). 8

b) Explain single and two address field sequencing techniques in micro programmed control unit. 7

a) Differentiate between Hardwired and Micro-programmed control unit. Explain the logic of Hardwired unit. 7

b) What is memory hierarchy? Explain with reason why do we need multilevel memory hierarchy? 7

a) Explain various communication techniques for I/O devices. 8

b) Differentiate between various mapping techniques when main memory data is to be mapped into cache memory. 7

a) What is power efficient processor? Explain dual core processor with respect to quad core processor. 8

b) Describe in brief about Flynn's Taxonomy. Also explain how parallelism can be experienced in uniprocessor systems. 7

Write short notes on: **(Any two)** 2×5

a) RISC vs. CISC

b). Vector processors and Array processors

c) BCD Adder

POKHARA UNIVERSITY

Semester : Spring

Level: Bachelor
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Course: Computer Architecture

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Differentiate between Computer Organization and Computer Architecture. Explain the functional view of the computer with reference to each component. 8

b) Define RTL. Explain about Bus and Register transfers in RTL. 7

a) Explain different types of registers used in Computer System with examples. What is the use of registers in Computer System? 7

b) Perform -15 *6 using Booth's algorithm. 8

a) Explain the restoring division algorithm with flowchart. 8

b) Explain the working principle of Micro-programmed implementation of control unit. Why is it slower than Hardwired Control Unit? 7

a) What is associative memory? How read /write operation takes place in associative memory? 7

b) As we know cache memory size is smaller than main memory, how is it possible to map main memory data into cache memory? Explain in detail. 8

a) How Interrupt driven I/O differ from programmed I/O? Explain with necessary diagram. 7

b) How instruction pipelining supports in parallelism? Explain. 8

a) Cache coherence problem arise in multiprocessor system and must be resolved. Justify this statement. 8

b) What are the performance issues that arise in multicore computers? 7

Write short notes on: (Any two) 2×5

a) Control unit.

b) RISC VS CISC

c) Flynn's Classification